

*Convolutional Neural Networks (CNNs) are a specialized class of deep neural networks predominantly used for analyzing visual imagery. They are highly effective for image classification, object detection, and facial recognition because they automatically and adaptively learn spatial hierarchies of features from input images.*

*A standard CNN architecture consists of three main types of layers: Convolutional layers, Pooling layers, and Fully Connected layers.*

*The Convolutional layer is the core building block. It applies a set of learnable filters (or kernels) across the input image. As the filter slides across the image, it computes dot products, resulting in a feature map that detects specific patterns like edges or textures. Next, the Pooling layer (often Max Pooling) reduces the spatial dimensions of the feature map. This downsampling decreases the computational load and helps make the model invariant to small translations or distortions in the image. Finally, the Fully Connected layers flatten the high-level features extracted by the previous layers and use them to output a final probability distribution over the target classes.*